

Sarah Alamdari

email: sarahhaham@gmail.com | website: <https://www.sarahahamdari.com/>
twitter: [@sarahahamdari](https://twitter.com/sarahahamdari) | github: <https://github.com/sarahahamdari>
google scholar: <https://scholar.google.com/citations?user=AV7XVKMAAAAJ&hl=en>

Summary

- My broader research interests and expertise lie at the intersection of **biology**, **deep learning**, and **molecular simulation**, with the goal of translating my expertise into useful scientific discoveries.
- I am motivated to better understand how data, scale, and ML can help advance our understanding of these biomolecules. Accordingly:
 - I was one of the first people to bring discrete diffusion to protein sequence design (**Alamdari** et al., *BioRxiv* 2023)
 - I later took these same concepts to build generative and predictive frameworks applied to protease systems for early cancer detection (Alonso and **Alamdari** et al., *Nature Communications*, 2026)
 - Then recently expanded this framework to understand the role of data composition and size, ultimately releasing the Dayhoff Atlas (Yang, **Alamdari** and Lee, et al., *BioRxiv*, 2026)
- I thrive in collaborative environments where I need to work closely with a team to solve large and challenging problems.

Experience

Dec 2023 – Present	Senior Applied Scientist, Biomedical ML Group, <i>Microsoft Research</i>
Apr 2022 – Dec 2023	Data Scientist, Biomedical ML Group, <i>Microsoft Research</i>
Jan 2022 – Apr 2022	Postdoctoral Researcher, Pfaendtner Research Group, <i>University of Washington</i>

Education

2021	PhD Chemical Engineering, <i>University of Washington</i> • Thesis: Discovery of Biomolecular Structure-Function Mechanisms with Computational Frameworks at the Nanoscale (Advisor: Jim Pfaendtner) • Completed <i>Data Science Option</i> certification
2016	BS Chemical Engineering, <i>Arizona State University</i> • Graduated <i>Magna Cum Laude</i>

Publications

First author contribution; * co-first authors; ‡ co-corresponding authors; + mentee

Microsoft Research

1. Kosonocky, W. C.⁺, **Alamdari, S.**, Yang, K. K., Amini, A. P. Closing the loop: Experimentally validated methods in artificial intelligence-driven protein design. *Current Opinion in Structural Biology*. **2026**.
2. Didi, K., **Alamdari, S.**, Lu, X. A., Wittman, B., Johnston, K. E., Amini, A., Madani, A., Czeneszew, M., Dallago, C.[‡], Yang, K. K.[‡], FLIP2: Expanding Protein Fitness Landscape Benchmarks for Real-World Machine Learning Applications. *International Conference on Machine Learning*, **2026**.

- **Accepted (~ top 3% of papers selected) and invited for an oral presentation (168 out of 23,918 submissions)**
3. Martin-Alonso, C.*, **Alamdari, S.***, Samad, T., Yang, K. K., Bhatia, S. N.†, and Amini, A. P.‡, Deep learning guided design of protease substrates, *Nature Communications*, **2026**.
 - **Highlighted in [MIT News](#), “AI-generated sensors open new paths for early cancer detection”**
 4. Yang, K. K.*, **Alamdari, S.***, Lee, A.*, Kaymak-Loveless K.†, Char, S., Bixi, G.†, Domingo-Enrich C., Wang, C.†, Lyu, S.†, Fusi, N., Tenenholtz, N.†, and Amini, A. P.‡, Dayhoff: An atlas of protein models and datasets. *BioRxiv*, **2026**. (*Under review at Cell Systems*)
 5. Char, S.*, Corely, N.*, **Alamdari, S.**, Yang, K. K. and Amini, A. P., ProtNote: a multimodal method for protein–function annotation. *Bioinformatics*, **2025**.
 6. Wang, C.†, **Alamdari, S.**, Domingo-Enrich C., Amini, A. P., and Yang, K. K., Towards deep learning sequence-structure co-generation for protein design. *Current Opinion in Structural Biology*. **2025**.
 7. Wu, K. E., Yang, K. K., van den Berg, R., **Alamdari, S.**, Zou, J. Y., Lu, A. X., and Amini, A. P., Protein structure generation via folding diffusion. *Nature Communications*. **2024**.
 8. **Alamdari, S.**, Thakkar, N.†, van den Berg, R., Tenenholtz, N., Strome, R. Moses, A. M., Lu, A. X., Fusi, N., Amini, A. P.‡, and Yang, K. K.‡, Protein generation with evolutionary diffusion: sequence is all you need. *eLife*, **2023**.
 - **Featured in [Tech Crunch](#) “Microsoft open sources EvoDiff, a novel protein-generating AI”**

University of Washington

9. **Alamdari, S.** and Pfaendtner J., Origins of Conformational Heterogeneity in Peptoid Helices Formed by Chiral N-1-Phenylethyl Side Chains. *Journal of Physical Chemistry B*, **2023**.
10. **Alamdari, S.**, Torkelson, K., Wang, X.†, Chen, C. L., Ferguson, A. L. and Pfaendtner J., Thermodynamic Basis for Stabilization of Helical Peptoids by Chiral Sidechains. *Journal of Physical Chemistry B*, **2023**.
11. Zhao, M., Zhang, S., Zheng, R., **Alamdari, S.**, Mundy, C. J., Pfaendtner, J., Pozzo, L. D., Chen C. L., DeYoreo, J. and Ferguson A. L., Computational and Experimental Determination of the Properties, Structure, and Stability of Peptoid Nanosheets and Nanotubes. *Biomacromolecules*, **2023**.
12. Strunge, K., Hoinkis, N., Lutz, H., **Alamdari, S.**, Roeters, S. J., Lu, H., Pfaendtner, J. and Weidner, T., Peptide Mimic of the Marine Sponge Protein Silicatein Fabricates Ultrathin Nanosheets of Silicon Dioxide and Titanium Dioxide. *Langmuir*, **2022**.
13. Zhao, M., Lachowski, K. J., Zhang, S., **Alamdari, S.**, Sampath, J., Mu, P., Mundy, C. J., Pfaendtner, J., De Yoreo, J. J., Chen, C. L. and Pozzo, L. D., Hierarchical Self-Assembly Pathways of Peptoid Helices and Sheets. *Biomacromolecules*, **2022**.
14. Roeters, S. J.*, Golbek, T. W.*, Bregnhøj, M.*, Drace, T., **Alamdari, S.**, Roseboom, W., Kramer, G., Šantl-Temkiv, T., Finster, K., Pfaendtner, J. and Woutersen, S., Ice-nucleating proteins are activated by low temperatures to control the structure of interfacial water. *Nature Communications*, **2021**.
15. Summers, S. R.*, **Alamdari, S.***, Kraft, C. J., Brunecky, R., Pfaendtner, J. and Kaar, J. L., Substitution of distal and active site residues reduces product inhibition of E1 from *Acidothermus Cellulolyticus*. *Protein Engineering, Design and Selection*, **2021**.

16. **Alamdari, S.***, Sampath, J.*, Prakash, A., Gibson, L. D. and Pfaendtner, J., Efficient Sampling of High-Dimensional Free Energy Landscapes: A Review of Parallel Bias Metadynamics. *Foundations of Molecular Modeling and Simulation*, **2021**.
17. Thompson, N. L., Cohen, T. A., **Alamdari, S.**, Hsu, C. W., Williamson, G.A. and Holmberg, V. C., DiffCapAnalyzer: A Python Package for Quantitative Analysis of Total Differential Capacity Data. *Journal of Open Source Software*, **2020**.
18. **Alamdari, S.***, Roeters, S. J.*, Golbek, T. W., Schmüser, L., Weidner, T.‡ and Pfaendtner, J.‡, Orientation and conformation of proteins at the air–water interface determined from integrative molecular dynamics simulations and sum frequency generation spectroscopy. *Langmuir*, **2020**.
19. Sampath, J.*, **Alamdari, S.***, and Pfaendtner, J., Closing the gap between modeling and experiments in the self-assembly of biomolecules at interfaces and in solution. *Chemistry of Materials*, **2020**.
20. Zhao, M., Sampath, J., **Alamdari, S.**, Shen, G., Chen, C. L., Mundy, C. J., Pfaendtner, J. and Ferguson, A.L., MARTINI-compatible coarse-grained model for the mesoscale simulation of peptoids. *The Journal of Physical Chemistry B*, **2020**.
21. Summers, S., Kraft, C., **Alamdari, S.**, Pfaendtner, J. and Kaar, J. L., Enhanced activity and stability of *Acidothermus cellulolyticus* endoglucanase 1 in ionic liquids via engineering active site residues and non-native disulfide bridges. *ACS Sustainable Chemistry and Engineering*, **2020**.
22. Hellner, B., **Alamdari, S.**, Pyles, H., Zhang, S., Prakash, A., Sprenger, K. G., De Yoreo, J. J., Baker, D., Pfaendtner, J.‡ and Baneyx, F.‡, Sequence–structure–binding relationships reveal adhesion behavior of the Car9 solid-binding peptide: an integrated experimental and simulation study. *Journal of the American Chemical Society*, **2020**.
23. **Alamdari, S.** and Pfaendtner, J., Impact of glutamate carboxylation in the adsorption of the α -1 domain of osteocalcin to hydroxyapatite and titania. *Molecular systems design and engineering*, **2020**.
24. Verreault, D., **Alamdari, S.**, Roeters, S. J., Pandey, R., Pfaendtner, J. and Weidner, T., Ice-binding site of surface-bound type III antifreeze protein partially decoupled from water. *Physical Chemistry Chemical Physics*, **2018**.

Arizona State University

25. Wang, G., Robert, C., Suslu, A., Chen, B., Yang, S., **Alamdari, S.**, Gerber, I. C., Amand, T., Marie, X., Tongay, S.‡ and Urbaszek, B.‡, Spin-orbit engineering in transition metal dichalcogenide alloy monolayers. *Nature communications*, **2015**.

Workshop Papers

*Presenting author; * co-first authors; ‡ co-corresponding authors; + mentee*

1. Yang, K. K.*, Alamdari, S.*, Lee, A.*, Kaymak-Loveless K., Char, S., Bixi, G., Domingo-Enrich C., Wang, C., Lyu, S., Fusi, N., Tenenholtz, N.‡, and Amini, A. P.‡, The Dayhoff Atlas: scaling sequence diversity for improved protein generation, *Machine Learning for Structural Biology (MLSB) Workshop (San Diego, CA) 2025. (poster)*
2. Yang, K. K.*, Alamdari, S.*, Lee, A.*, Kaymak-Loveless K., Char, S., Bixi, G., Domingo-Enrich C., Wang, C., Lyu, S., Fusi, N., Tenenholtz, N.‡, and Amini, A. P.‡, The Dayhoff Atlas: scaling sequence diversity for improved protein generation, *The Molecular Machine Learning (MoML) Conference @ MIT (Cambridge, MA) 2025. (selected as “Top 4 papers” spotlight talk)*

3. Lee, A., Alamdari, S., Yang, K. K., Amini, A. P. Augmenting protein language models with synthetic structures. *ICLR Generative and Experimental perspectives in bioMolecular design (GEM) Workshop. (Singapore). 2025. (poster, selected for direct submission at Nature Biotechnology)*
4. Alamdari, S., Thakkar, N., van den Berg, R., Lu, A. X., Fusi, N., Amini, A. P. and Yang, K. K. Protein generation with evolutionary diffusion. *NeurIPS Generative AI and Biology (GenBio) Workshop. (New Orleans, LA). 2023. (poster)*
5. Alamdari, S., Thakkar, N., van den Berg, R., Lu, A. X., Fusi, N., Amini, A. P. and Yang, K. K. Protein generation with evolutionary diffusion. *NeurIPS Machine Learning in Structural Biology (MLSB) Workshop. (New Orleans, LA) 2023. (selected as a spotlight talk)*
6. Alamdari, S., Thakkar, N., van den Berg, R., Lu, A. X., Fusi, N., Amini, A. P. and Yang, K. K. Protein generation with evolutionary diffusion. *The Molecular Machine Learning (MoML) Conference @ MIT. (Cambridge, MA) 2023. (poster)*

Open Source Software

FLIP2, Contributor, <https://github.com/J-SNACKKB/FLIP>

Dayhoff, Primary Developer, <https://github.com/microsoft/dayhoff>

CleaveNet, Primary Developer, <https://github.com/microsoft/cleavenet>

EvoDiff, Primary Developer, <https://github.com/microsoft/evodiff/>

Select Invited Talks

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| 2026 | Sequence based approached to protein design. <i>Protein Talk Protein Evolution, Design and Informatics, Biochemical Society (Edinburgh, Scotland)</i> |
| 2025 | Sequence based approaches to protein design. <i>AI4Science Seminar, Chalmers University (Gothenburg, Sweden)</i> |
| 2025 | Protein design in sequence space. <i>39th Annual Symposium of The Protein Society (San Francisco, CA – delivered virtually)</i> |
| 2025 | Generative sequence-based approaches to protein design. <i>Schrödinger Biologics Symposium (Cambridge, MA)</i> |
| 2025 | EvoDiff: A General-Purpose Diffusion Framework for Programmable, Sequence-first Protein Design. <i>Applied ML Days, EPFL (Lausanne, Switzerland)</i> |
| 2024 | Discrete diffusion models for generative protein design. <i>Boston Protein Design and Modeling Club (BPDMC) (Cambridge, MA)</i> |
| 2024 | In-silico design of protease-cleavable peptide substrates, <i>Gordon Research Conference on Proteolytic Enzymes and Their Inhibitors (Lucca, Italy)</i> |
| 2024 | AI Driven Revolutions in Protein Engineering: Pioneering the Next Wave of Biotech Innovations, <i>SynBioBeta (San Jose, CA)</i> |
| 2024 | Discrete diffusion models for controllable protein generation. <i>Molecular ML Reading Group, Rosetta Commons (virtual)</i> |
| 2023 | Discrete diffusion models for controllable protein generation. <i>ML Ideas, Microsoft Research (Cambridge, MA)</i> |
| 2023 | An introduction to diffusion models. <i>Models, Inference & Algorithms (MIA) Initiative Seminar Series at the Broad Institute (Cambridge, MA)</i> |
| 2023 | Sequence-first generative approaches for protein design. <i>AstraZeneca Journal Club. (virtual)</i> |

- 2023** Protein sequence generation with evolutionary diffusion. *Google Research. (Cambridge, MA)*
- 2022** Exploration of the Peptoid Folding Landscape with Metadynamics. *University at Buffalo CBE Seminar. (Buffalo, NY)*
- 2021** A Thermodynamic Basis for Peptoid Assembly into a Helix by A Chiral Sidechain. *Broadbelt Group, Northwestern. (Evanston, IL)*
- 2021** Simulations of the Peptoid Folding Landscape with Metadynamics. *UW ChemE Seminar. (Seattle, WA)*
- 2021** Exploration of the Peptoid Folding Landscape with Metadynamics. *Molecular Foundry Lawrence Berkeley National Lab. (virtual)*
- 2021** Combining simulation and experiment to study protein structure at interfaces. *Biophysics Society Student Networking Event. (virtual)*
- 2021** Biomolecular Assembly at Interfaces. *ChE Future Faculty Seminar Series. (virtual)*

Select Awards

- 2022** VITAL Future Faculty, *University at Buffalo*
- 2021** Chemical Engineering Faculty Lecture Award, *University of Washington*
- 2021** Chemical Computing Group Excellence Student Award, *ACS COMP*
- 2021** Graduate and Postdoc Women's Fellowship, *D.E. Shaw Research*
- 2020** Rising Star in Chemical Engineering, *MIT*
- 2020** Husky 100, *University of Washington*
- 2020** ACES Graduate Student Symposium Top Speaker Award, *University of Washington*
- 2020** COMSEF Graduate Student Award, *AIChE*
- 2019** NSF-AGEP Research Exchange, *The California Alliance*
- 2018** Graduate Research Fellowship Program (GRFP), *NSF*
- 2018** National Research Traineeship (NRT), *NSF*
- 2016** ARCS Fellow, *University of Washington*